

Some general properties of linear transformations (contd.)

Last lecture:

Theorem 1 Let $T : V \rightarrow W$ be a linear transformation.

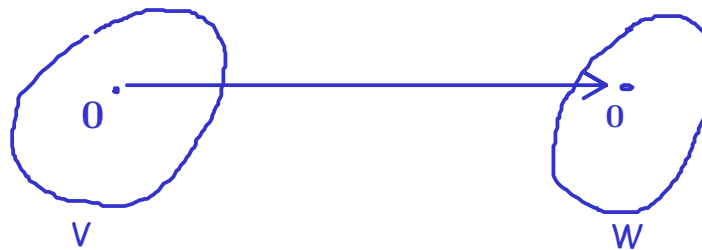
Then

1. $T(0) = 0$.
zero vector in V $\rightarrow$ zero vector in W
2. For any $\mathbf{v} \in V$,
 $T(-\mathbf{v}) = -T(\mathbf{v})$.

Application:

The theorem implies that any linear transformation maps the zero vector to the zero vector. Hence if a function does not have this property then it is not a linear transformation.

This is a nice first check that tells us if there is some chance that the function is a linear transformation.



Ex $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$

We have

$$T \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We can conclude that T is not a linear transformation.

Ex $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} x_1 + 2 \\ x_2 \end{pmatrix}$$

We have

$$T \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Hence T is not a linear transformation.

However, note:

$$T \text{ defined by } T \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} x_1^2 \\ x_2^2 \end{pmatrix}$$

maps the zero vector to the zero vector, but is not a linear transformation.

Theorem 2 Let $T : V \rightarrow W$ be a linear transformation,

$u_1, u_2, \dots, u_n \in V$, and $\lambda_1, \lambda_2, \dots, \lambda_n$ scalars.

Then

$$T(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n) = \lambda_1 T(\mathbf{u}_1) + \lambda_2 T(\mathbf{u}_2) + \dots + \lambda_n T(\mathbf{u}_n).$$

Proof This theorem just combines that linear transformations preserve addition and scalar multiplication (for the case of more than two vectors).

$$\begin{aligned} T(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n) &= T(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_{n-1} \mathbf{u}_{n-1}) + T(\lambda_n \mathbf{u}_n) \\ &\quad \vdots \quad \leftarrow \text{repeatedly use that } T \text{ preserves addition} \\ &= T(\lambda_1 \mathbf{u}_1) + T(\lambda_2 \mathbf{u}_2) + \dots + T(\lambda_n \mathbf{u}_n) \\ &= \lambda_1 T(\mathbf{u}_1) + \lambda_2 T(\mathbf{u}_2) + \dots + \lambda_n T(\mathbf{u}_n). \end{aligned}$$

Because T preserves
scalar multiplication



Theorem 3 Let $T : V \rightarrow W$ be a linear transformation and $\mathbf{u}, \mathbf{v} \in V$.

Then $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$.

Proof

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u} + (-\mathbf{v})) = T(\mathbf{u}) + T(-\mathbf{v}) = T(\mathbf{u}) + (-T(\mathbf{v})) = T(\mathbf{u}) - T(\mathbf{v}).$$

by a theorem from the
last lecture



Important remark: Let $T : V \rightarrow W$ be a linear transformation and $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ a basis for V . Then T is uniquely determined by $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$.
That is, it is enough to know where the basis vectors are mapped to.

Why?

Let $\mathbf{v}$ be any vector in V .

Because $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is a basis, we can write $\mathbf{v}$ as

$$\mathbf{v} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_n \mathbf{b}_n$$

for suitable scalars $\lambda_1, \lambda_2, \dots, \lambda_n$.

Because T is linear, we obtain

$$T(\mathbf{v}) = \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_n T(\mathbf{b}_n).$$

Hence it is enough to know $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$.

Example Let $T : P_3 \rightarrow P_2$ be a linear transformation.

We know that $\{1, x, x^2, x^3\}$ is a basis for P_3 .

We are given where the basis vectors are mapped to by T :

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2x, \quad T(x^3) = 4x^2$$

Then we know what T maps to for any $p \in P_3$:

$$\begin{aligned} T(a \cdot 1 + bx + cx^2 + dx^3) &= a \underbrace{T(1)}_0 + b \underbrace{T(x)}_1 + c \underbrace{T(x^2)}_{2x} + d \underbrace{T(x^3)}_{4x^2} \\ &= b + 2cx + 4dx^2 \end{aligned}$$

Kernel and range of a linear transformation

a.) Kernel

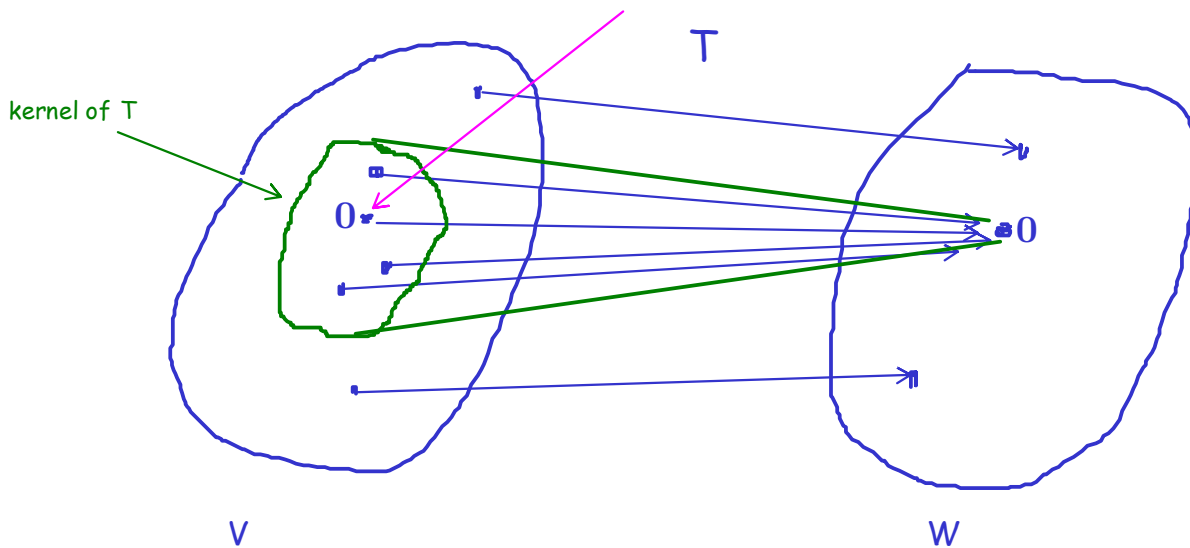
Definition Let $T : V \rightarrow W$ be a linear transformation.

Then

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}.$$

kernel of T

The zero vector is always mapped to the zero vector. But in general, there are other vectors that also get mapped to the zero vector. The set of these vectors is called the kernel.



Example The linear transformation that sends everything to the zero vector:

$$T : V \rightarrow W \text{ such that for all } \mathbf{v} \in V, T(\mathbf{v}) = \mathbf{0}.$$

Check that this indeed is a linear transformation!

Then $\ker(T) = V$. largest possible kernel

Example $T : V \rightarrow V$, where for all $\mathbf{v} \in V$, $T(\mathbf{v}) = \mathbf{v}$. (Identity map)

Check that this indeed is a linear transformation!

Then $\ker(T) = \{\mathbf{0}\}$. smallest possible kernel (the zero vector is always there!)

Example Let $V = P_3$ and $W = P_2$.

Define the function $T : P_3 \rightarrow P_2$ by

$$T(p(x)) = p'(x).$$

Then

$\ker(T)$ = The set of all constant functions

Example $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Recall that this is a rotation by 90 degrees.

$$\begin{aligned} \ker(T) &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \\ &= \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \end{aligned}$$

We can also see this geometrically because T is a rotation.

We see that the kernel equals the null space of the matrix used to define the linear transformation.

This holds more generally:

Fact Let $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ and T_A the linear transformation defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Then $\ker(T_A) = NS(A)$.

By definition, $\ker(T)$ is a subset of V . Now we will see that $\ker(T)$ is not just an arbitrary subset of V , but even a subspace of V .

Theorem If $T : V \rightarrow W$ is a linear transformation then $\ker(T)$ is a subspace of V .

Proof

1.) $\ker(T)$ is non-empty because we know that $\mathbf{0} \in \ker(T)$.

2.) $\ker(T)$ is closed under addition:

Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be arbitrary vectors in $\ker(T)$.

$$\text{Then } T(\mathbf{v}_1 + \mathbf{v}_2) = \underbrace{T(\mathbf{v}_1)}_{\mathbf{0}} + \underbrace{T(\mathbf{v}_2)}_{\mathbf{0}} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Hence $\mathbf{v}_1 + \mathbf{v}_2 \in \ker(T)$.

3.) $\ker(T)$ is closed under scalar multiplication:

Let $\mathbf{v} \in \ker(T)$ and λ any scalar.

$$\text{Then } T(\lambda\mathbf{v}) = \lambda \underbrace{T(\mathbf{v})}_{\mathbf{0}} = \lambda \cdot \mathbf{0} = \mathbf{0}.$$

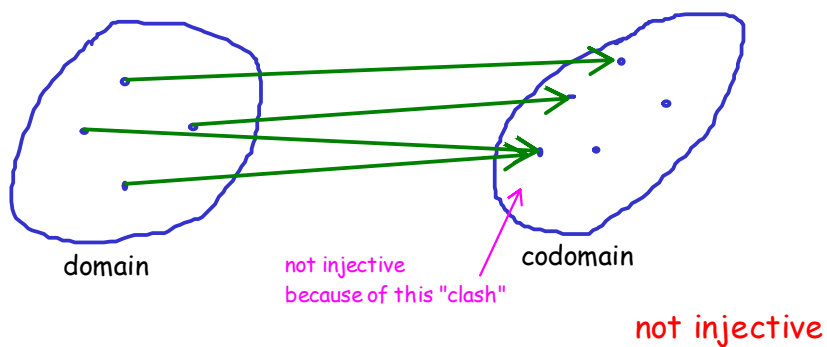
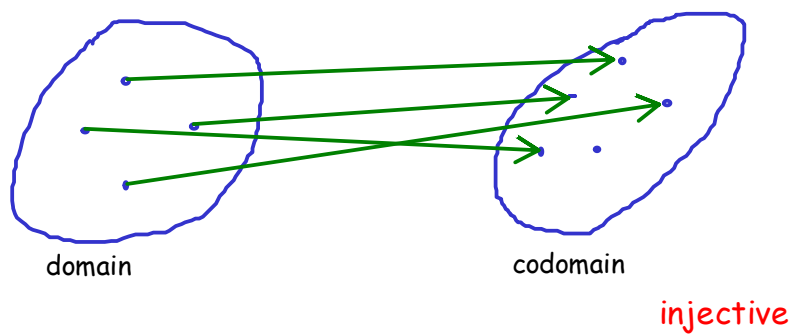
Hence $\lambda\mathbf{v} \in \ker(T)$.



Injective linear transformations

Definition A function $f : A \rightarrow B$ is injective (= one-to-one) if $f(x) = f(y)$ implies $x = y$

Contrapositive: $x \neq y$ implies $f(x) \neq f(y)$.



Injective = "no clashes" (every element in the codomain gets mapped to by at most one element in the domain)

The following theorem gives us a nice way to determine if a given linear transformation is injective:

Theorem 5 A linear transformation $T : V \rightarrow W$ is injective if and only if $\ker(T) = \{0\}$

Proof

” $\implies$ ”

Suppose T is injective.

The idea is: We know that the zero vector is mapped to the zero vector. But because T is injective, nothing else can be mapped to the zero vector. Hence zero vector is the only vector mapped to the zero vector, which means the kernel consists only of the zero vector.

More formally:

Let $v \in \ker(T)$.

Then $T(v) = 0$ because v is in the kernel.

and $T(0) = 0$ because 0 is always mapped to 0 .

Hence $T(v) = T(0)$.

Because T is injective, it follows that $v = 0$.

Hence the kernel only consists of the zero vector.

” $\impliedby$ ”

Suppose $\ker(T) = \{0\}$.

Let v_1 and v_2 be arbitrary vectors in V such that $T(v_1) = T(v_2)$.

Then $T(v_1) - T(v_2) = 0$

By Theorem 3, $T(v_1 - v_2) = 0$.

Because $\ker(T) = \{0\}$, this implies that $v_1 - v_2 = 0$.

Hence $v_1 = v_2$.

We therefore have shown that T is injective.



Example $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Recall that this is a rotation by 90 degrees.

We have already seen that $\ker(T) = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$, which tells us that T is injective.

This makes sense because T is a rotation.

Example $T_B : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T_B \left(\underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\mathbf{x}} \right) = \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\mathbf{x}}$$

We can see that T_B is a projection on the y-axis.

We can determine that

$$\ker(T_B) = \left\{ \begin{pmatrix} x_1 \\ 0 \end{pmatrix} : x_1 \in \mathbb{R} \right\} \neq \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

The kernel consists of the whole x-axis. (The x-axis is projected to the zero vector.)

Hence Theorem 5 tells us that T_B is not injective.

This makes sense because T_B is a projection.